PHYS 206 Experiment 6 Lab Report: Determination of e/m

Ahmet Buğra Ertürk
Lab Partner(s): Sermet Arda Topal

May 25, 2026

1 Introduction

The e/m Apparatus provides a simple method for measuring e/m, the charge to mass ratio of the electron. The method is similar to that used by J.J. Thomson in 1897. A beam of electrons is accelerated through a known potential, so the velocity of the electrons is known. A pair of Helmholtz coils produces a uniform and measurable magnetic field at right angles to the electron beam. This magnetic field deflects the electron beam in a circular path. By measuring the accelerating potential (V), the current to the Helmholtz coils (I), and the radius of the circular path of the electron beam (r), e/m is easily calculated:

$$e/m = \frac{2V}{B^2r^2}$$

The e/m apparatus also has deflection plates that can be used to demonstrate the effect of an electric field on the electron beam. This can be used as a confirmation of the negative charge of the electron, and also to demonstrate how an oscilloscope works. A unique feature of the e/m tube is that the socket rotates, allowing the electron beam to be oriented at any angle (from 0-90 degrees) with respect to the magnetic field from the Helmholtz coils. You can therefore rotate the tube and examine the vector nature of the magnetic forces on moving charged particles.

2 Theory and Analysis of e/m Measurement

The magnetic force (F_m) acting on a charged particle of charge q moving with velocity v in a magnetic field (B) is given by the equation $F_m = qv \times B$, (where F, v, and B are vectors and $\times$ is a vector cross product). Since the electron beam in this experiment is perpendicular to the magnetic field, the equation can be written in scalar form as:

$$F_m = evB \tag{1}$$

where e is the charge of the electron. Since the electrons are moving in a circle, they must be experiencing a centripetal force of magnitude:

$$F_c = mv^2/r \tag{2}$$

where m is the mass of the electron, v is its velocity, and r is the radius of the circular motion. Since the only force acting on the electrons is that caused by the magnetic field, $F_m = F_c$, so equations 1 and 2 can be combined to give:

$$evB = mv^2/r \implies e/m = v/Br \tag{3}$$

Therefore, in order to determine e/m it is only necessary to know the velocity of the electrons, the magnetic field produced by the Helmholtz coils, and the radius of the electron beam.

The electrons are accelerated through the accelerating potential V, gaining kinetic energy equal to their charge times the accelerating potential. Therefore $eV = 1/2mv^2$ and the velocity of the electrons is:

$$v = (2eV/m)^{1/2} \tag{4}$$

The magnetic field produced near the axis of a pair of Helmholtz coils is given by the equation:

$$B = \frac{[N\mu_0]I}{(5/4)^{3/2}a} \tag{5}$$

A derivation for this formula can be found in most introductory texts on electricity and magnetism. Equations 4 and 5 can be plugged into equation 3 to get a final formula for e/m:

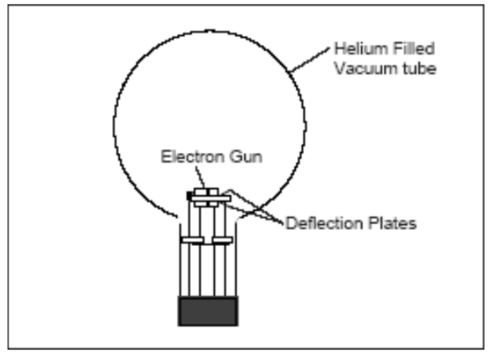
$$e/m = \frac{2V(5/4)^3a^2}{(N\mu_0 Ir)^2} \tag{6}$$

where V is the accelerating potential, a (= 15 cm = 0.15 m) is the radius of the Helmholtz coils, N is the number of turns on each Helmholtz coil (= 130), μ_0 is the permeability constant ($= 4\pi \times 10^{-7}$), I is the current through the Helmholtz coils, and r is the radius of the electron beam path.

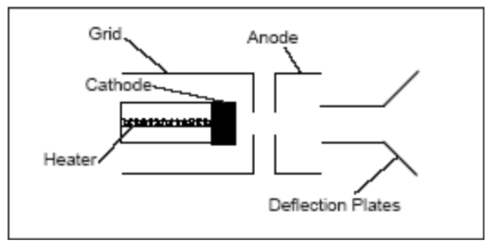
3 Equipment Needed

- **The e/m Tube:** Filled with helium at a pressure of 10^{-2} mm Hg, and contains an electron gun and deflection plates. The electron beam leaves a visible trail in the tube, because some of the electrons collide with helium atoms, which are excited and then radiate visible light.
- **Electron Gun:** The heater heats the cathode, which emits electrons. The electrons are accelerated by a potential applied between the cathode and the anode. The grid is held positive with respect to the cathode and negative with respect to the anode, which helps to focus the electron beam.
- **The Helmholtz Coils:** The geometry of Helmholtz coils—the radius of the coils is equal to their separation—provides a highly uniform magnetic field. The coils have a radius and separation of 15 cm, and each coil has 130 turns. The magnetic field produced is proportional to the current through the coils: $B(\text{tesla}) = (7.80 \times 10^{-4})I$.
- **Controls and Accessories:** The control panel of the e/m apparatus has all connections labeled. A cloth hood can be placed over the top of the apparatus so the experiment can be performed in a lighted room. A mirrored scale is attached to the back of the rear Helmholtz coil to measure the radius of the beam path without parallax error.

(a) e/m Apparatus



(b) e/m Tube



(c) Electron Gun

Figure 1: Equipment Setup for the Experiment

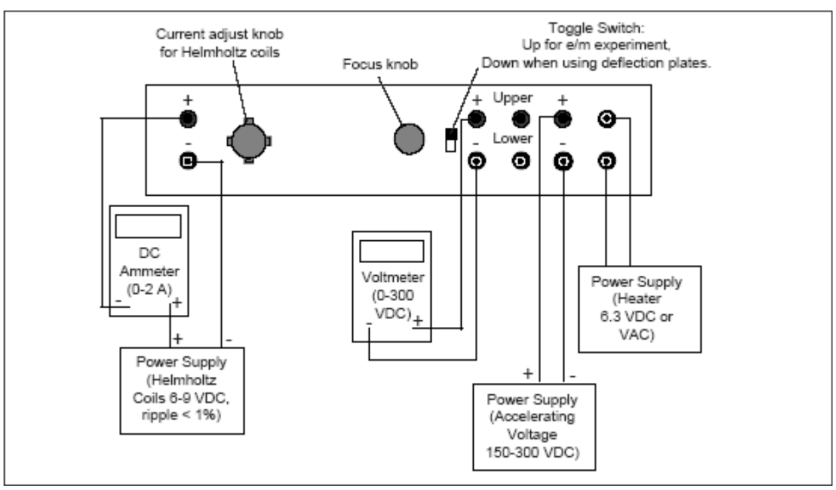


Figure 2: Connections for e/m Experiment

4 Experimental Procedure

4.1 Measuring e/m

1. If you will be working in a lighted room, place the hood over the e/m apparatus.

2. Flip the toggle switch up to the e/m MEASURE position.

3. Turn the current adjust knob for the Helmholtz coils to the OFF position.

4. Connect your power supplies and meters to the front panel of the e/m apparatus.

5. Adjust the power supplies to the following levels:

   Heater at 6.3 VAC or VDC
   Electrodes at 150 to 300 VDC
   Helmholtz Coils at 6-9 VDC with a ripple of less than 1%.

   CAUTION: The voltage to the heater should NEVER exceed 6.3 volts, as higher voltages will burn out the filament and destroy the tube.

6. Slowly turn the current adjust knob for the Helmholtz coils clockwise, taking care that the current does not exceed 2 A.

7. Wait several minutes for the cathode to heat up. When it does, the electron beam will emerge and curve. Check that the electron beam is parallel to the Helmholtz coils; if it is not, turn the tube until it is without taking it out of its socket.

8. Carefully read the current to the Helmholtz coils (I) from your ammeter and the accelerating voltage (V) from your voltmeter.

   Current to Helmholtz coils = I = 1.36 A
   Accelerating voltage = V = 255 V

9. Measure the radius of the electron beam (r) by looking through the tube. Align the beam with its reflection on the mirrored scale to avoid parallax errors, measure on both sides, and average the results.

   Electron Beam Radius = r = 4.75 cm = 0.0475 m

4.2 Deflections of Electrons in an Electric Field

IMPORTANT: Do not leave the beam on for long periods of time in this mode, as it will ultimately wear through the glass walls of the tube.

1. Setup the equipment as described for measuring e/m, except flip the toggle switch to ELECTRICAL DEFLECT.
2. Do not supply current to the Helmholtz coils. Connect a 0-50 VDC power supply between the DEFLECT PLATES (UPPER and LOWER).
3. Apply the 6.3 VDC/VAC to the HEATER and 150-300 VDC to the ELECTRODES, and wait several minutes to warm up the cathode.
4. Slowly increase the voltage to the deflection plates from 0 V to approximately 50 VDC. Note that the beam is bent towards the positively charged plate.

5 Data Collection and Analysis

Using the values from the experiment:

- Accelerating potential, $V = 255$ V
- Helmholtz coil current, $I = 1.36$ A
- Radius of electron beam, $r = 0.0475$ m
- Radius of coils, $a = 0.15$ m
- Number of turns, $N = 130$
- Permeability constant, $\mu_0 = 4\pi \times 10^{-7}$ T · m/A

First, we calculate the magnitude of the magnetic field B produced by the Helmholtz coils using Equation 5:

$$B = \frac{130 \cdot (4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}) \cdot 1.36\ \mathrm{A}}{(5/4)^{3/2} \cdot 0.15\ \mathrm{m}} \approx 1.060 \times 10^{-3}\ \mathrm{T}$$

Next, substituting B, V, and r into Equation 3 to find the experimental e/m:

$$e/m = \frac{2(255\ \mathrm{V})}{(1.060 \times 10^{-3}\ \mathrm{T})^2 \cdot (0.0475\ \mathrm{m})^2} \approx 2.012 \times 10^{11}\ \mathrm{C/kg}$$

The accepted theoretical value for e/m is approximately 1.759×10^{11} C/kg. The percent error of our measurement is calculated as:

$$\% \text{ Error} = \frac{|2.012 \times 10^{11} - 1.759 \times 10^{11}|}{1.759 \times 10^{11}} \times 100 \approx 14.4\%$$

6 Conclusion and Discussion

In this experiment, e/m was measured to be 2.012×10^{11} C/kg, comparing to the accepted theoretical value of 1.759×10^{11} C/kg, this result leads to 14.4% error. This error may be caused by the measurement methods (especially radius of electron beam) and instrumentation errors.

The most important source of error in this experiment is the velocity of the electrons. Electrons collide with helium atoms in the tube, and these collisions reduce their kinetic energy and velocity even more. In the formula, e/m is proportional to $1/r^2$. Since a lower velocity causes a smaller radius r, the actual radius we measure becomes smaller than it should be. Measuring a smaller radius makes our experimental e/m value higher than the theoretical value. However, the experiment still clearly shows how electrons deflect in a magnetic field.